

AYMANE AIT DADS

Intern Software Engineer - Edge AI Engineering | Model Deployment · LoRA Fine-Tuning · ML Pipeline Engineering

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PROFESSIONAL SUMMARY

Ranked **17th out of 3,000+ teams** in the NVIDIA Nemotron Reasoning Challenge by designing and executing an end-to-end LoRA SFT pipeline on a 30B-parameter model, demonstrating production-grade AI engineering at scale. Reduced manual network analysis time by **60%** at Orange Maroc by deploying automated ML pipelines on 100K+ records, translating model outputs directly into operational recommendations for network teams. Brings hands-on experience in AI model training, edge-relevant inference optimization (BF16, vLLM, grad_accum), and hardware-aware pipeline design using PyTorch, Hugging Face Transformers, and PEFT - precisely the toolkit required to push AI deployment performance at Roofline. Aims to contribute to simplifying AI deployment at the edge by applying ablation-driven engineering discipline and deep knowledge of model optimization to diverse hardware targets.

CORE COMPETENCIES

AI Model Deployment | Edge AI Engineering | Model Optimization | ML Pipeline Engineering | Hardware-Aware Inference | LoRA Fine-Tuning | Ablation Studies | Mixed-Precision Training

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and deployed an end-to-end **ML pipeline** combining Random Forest and K-Means on **100K+ fiber-optic network records** to classify network quality across thousands of nodes.
- Built a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible Pandas/Scikit-learn processing scripts.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, bridging data science results and operational engineering teams.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) targeting all attention projections, MLP layers, and lm_head.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · TRL · PEFT · vLLM · Hugging Face Transformers · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head, dual learning rates, and selective unfreezing of up to 8 transformer blocks.
- Built a **3-model ensemble with 10-view TTA** (flips, crops, blur, resizes) trained on 250K images across 25+ generator types in FP16 mixed precision; submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.

PyTorch · OpenCLIP · Hugging Face Transformers · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing, Image Security, Statistics, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Hugging Face Transformers | Unsloth | TRL | PEFT | Scikit-learn | vLLM

LLM & Model Optimization: LoRA | QLoRA | SFTTrainer | BF16/FP16 Mixed Precision | Completion-Only Loss | Mamba-Transformer | Chain-of-Thought

Data, Infra & Tooling: Python | SQL | Git | Docker | CUDA | Triton | Pandas | Power BI